

Ratification kit: the hand-verification gaps

What the scripts could not check, item by item, with references

Prepared end of session 16

2026-07-11

How to use this document. Every item below was verified by a fresh-context adversarial review with from-scratch machine receipts; each receipt honestly lists the steps that are *hand-proof-only* — infinite objects, topology, choice principles, and source-reading judgment calls that no finite script reaches. Those steps are where your eye is the only remaining verification layer. For each item: the claim chain, the receipt, then each gap with (a) exactly what to check, (b) where the written proof lives, (c) the primary reference with page, (d) what a failure would look like and what it would take down. Ratifying an item = you accept its gaps after inspection (or spot-check the ones you choose and accept the receipt for the rest); tell me and I mark the records.

Suggested order. Items 2–3–4 are serial (each uses the previous); item 1 is quick and independent; item 5 is a separate lane entirely. Item 2’s gaps are the highest-leverage in the corpus: A2 and T3 carry everything after them.

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1 Item 1 (s9): the MPT theorem-let — no non-Boolean concrete σ -class OML has all σ -additive states Jauch–Piron

Claim chain (attack note §7c(i)): on a concrete σ -class, every Dirac state is two-valued and σ -additive [glue, [[HAND]]; all-Dirac-states-JP $\Leftrightarrow C_{\text{card}}$ [MPT Prop 3.2, *their* proof]; $C_{\text{card}} \Rightarrow$

downward-directed [MPT p. 10, one direction only]; downward-directed + lattice $\Rightarrow$ Boolean [MPT Prop 4.4, *their* proof]. Consequences: BNPW 1985 excluded from $\text{Adm} \cap \text{OML}$ under every reading (disjunction); Skeleton C first half refuted by MO_2 .

Receipt: §7c(iii-b)/(iii-c) of `../oml_attack/oml_lattice_regularity_attack.md`; `../verification/mo2_jp-check.p` (seven properties). Reviewer read the MPT PDF in full.

Gap 1.1. *The glue: Dirac states are σ -additive on σ -classes.* Three lines: if $E = \biguplus_n E_n \in L$ (disjoint union), then x lies in exactly one E_n , so $\delta_x(E) = \sum_n \delta_x(E_n)$.

What to look for: the only subtlety is that σ -class σ -completeness is the *union* sense (disjoint countable unions are set unions and lie in L), matching `papers/sigma_essential/sigma_essential_body.tex` q:bare clause (b). Confirm the theorem-let’s scope line: “Adm’s σ -completeness IS the σ -class sense.”

Red flag: if Adm’s completeness were *sup*-based (suprema exist but are not set unions), the glue fails — that is exactly the distinction MPT Prop 4.6 (p. 11) navigates, and the reviewer checked the senses match. Reference: Müller–Pták–Tkadlec, *Int. J. Theor. Phys.* 31 (1992) 843–854, PDF `muller_ptak_tkadlec.1992.covering.properties.pdf`: Def 1.1 (p. 2, *finite* closure only — their default logic is weaker than a σ -class), Prop 3.2 (p. 6, no side conditions), Prop 4.4 (p. 10).

Gap 1.2. *Direction of the Prop 3.2 bridge.* Prop 3.2’s C_{card} is per-point ($\forall x \in A \cap B \exists C \ni x, C \subseteq A \cap B$), which implies downward-directedness strictly — it is NOT an iff.

What to look for: §7c(i) uses only the correct direction ($\Rightarrow$); check no later use silently upgrades it.

Gap 1.3. *The BNPW disjunction* (§7c, “Consequence for BNPW”). Pure logic over an under-determined source: either BNPW’s object is a σ -class (then not a lattice, by the theorem-let) or it is not σ -closed (then not an Adm-style carrier).

What to look for: the disjunction exhausts the readings of MPT’s Q5.4 note (p. 12); the Bunce–Hamhalter 2000 ILL pull remains confirmation-grade only — nothing is cited from it.

Red flag: any future text citing BNPW as a lattice witness or non-witness *simpliciter*.

2 Item 2 (s11 math + s12 edits): attack note §9 — the foundation layer (A1, L0, A2, T1, T3, P1, Conjecture B')

Why this item matters most: A2 (blocks are σ -fields) and T3 (Dirac realization on countably generated blocks) are load-bearing for every subsequent session (§10’s factoring, §11’s $P^=$ and ladder). A defect here propagates everywhere.

Receipt: `../oml_attack/PROOF_READ.2026-07-10_attack.s9.md` (118/118 checks, three flagged high-risk steps cleared); scripts `../verification/proof_read.2026-07-10_s11/`.

Lean bolster (2026-07-11, s17 — added after this kit was issued). `formalization/QuerySystem/QuerySystem` (builds; `#print axioms` receipts at file end) converts most of this item from hand-check to machine receipt:

- **Gap 2.4 DISCHARGED** — A1(a) Zorn extension (`exists_isMaxBlock_superset`), A1(b) closure including countable disjoint unions (`IsMaxBlock.iUnion_mem`), and A1(c) blockwise σ -reduction (`isSigmaOn_carrier_iff_maxBlocks`); axiom-free (classical base only).
- **Gap 2.5 DISCHARGED** — L0, both directions (`Compat.isGreatest_inter`, `compat_of_commuting_deco`); axiom-free.

- **Gap 2.2 DISCHARGED** — T3 in full, including the σ -continuity-from-above step and the dichotomy σ -subfield step (`dirac_realization_of_countablyGenerated`); axiom-free. Both hidden joints are settled by construction: the Lean T3 is stated on an abstract countably generated σ -field — independent of A2, so no circularity — and the dichotomy family’s closure under countable unions is proved (`dichotomy_of_generateMeasurable`).
- **Gap 2.1 REDUCED** — A2 (blocks are σ -fields: `IsMaxBlock.inter_mem`, `IsMaxBlock.toMeasurableSpace`) is machine-checked *modulo one flagged axiom* `foulis_holland_commute`: the Foulis–Holland/commutant step, stated as the commuting decomposition of $A \cap B$ along C ; the concrete upgrade through L0 is *proved*, not assumed. The remaining hand-check is exactly this item’s original *look-for*: read Bruns–Harding Props 2.2–2.8 / Kalmbach Thm 5 once and confirm the statement matches the axiom. Nothing else in 2.1 remains.
- **Gap 2.3 remains** (the two counterexample checks are not formalized), but note the positive T3 statement no longer rests on hand re-derivation, so (a)/(b) are corroboration of hypotheses being load-bearing, no longer proof-carrying.

The composite — σ -additive states are Dirac on countably generated blocks (`IsMaxBlock.dirac_realization`) — lists exactly [`propext`, `Classical.choice`, `Quot.sound`, `foulis_holland_commute`]. Ratifying item 2 now means: read the one axiom against its citation, and accept the build.

Gap 2.1. *A2’s infinite commutant case.* The step “every pairwise-compatible subset — of any cardinality — generates a Boolean subalgebra” rests on Bruns–Harding Prop 2.8, checked by the scout to be stated for *arbitrary* subsets (via commutant closure, not a directed-union limit). *What to look for*: read Bruns–Harding 2000, “Algebraic Aspects of Orthomodular Lattices” (in *Current Research in Operational Quantum Logic*, Kluwer), Props 2.2–2.4 (commutant $C(a)$ closed under $'$ and all existing joins/meets), Prop 2.8 (arbitrary pairwise-commuting subsets), Prop 3.1 (blocks). Verify Prop 2.8’s statement covers infinite sets with no separability hypothesis. Nuance already pinned in the receipt: Foulis–Holland (Kalmbach 1983, Thm 5 p. 25; Holland 1964 *TAMS* 112, **not** 1963) gives only distributivity of the generated *sublattice*; the Boolean *subalgebra* conclusion for infinite families is exactly what Prop 2.8 adds.

Red flag: if Prop 2.8 needed a finiteness or completeness hypothesis, A2’s “blocks are σ -fields” would lose the infinite case, and with it T3, $P^=$, and the §10/§11 superstructure.

Gap 2.2. *T3’s σ -continuity derivation* (infinite; the scripts only exercised finite σ -fields and the pentagon). Re-derive: F_n = intersection of the first n generators’ value-1 sides ($\in Bl$ by A2); $\nu(F_n) = 1$ by two-valued f.a. induction; $D = \bigcap_n F_n \in Bl$; $F_1 = D \uplus \biguplus_n (F_n \setminus F_{n+1})$ with each difference ν -null, so σ -additivity *on* Bl (supplied by A1c) gives $\nu(D) = 1$, hence $D \neq \emptyset$; finally $\{A \in Bl : D \subseteq A \text{ or } D \cap A = \emptyset\}$ is a σ -subfield containing the generators.

What to look for: two hidden joints: (a) “countable unions in Bl ARE set unions” — that is A2, so check there is no circularity (A2’s proof uses only Bruns–Harding + L0 + A1b, never T3 — confirm by reading §9c’s A2 paragraph); (b) the σ -subfield step needs closure under countable unions of the dichotomy family, one line but write it.

Red flag: a use of σ -continuity *from above* on L itself (not available in a σ -class); the proof must route through the block, where A2 makes it a genuine σ -field.

Gap 2.3. *T3’s load-bearing counterexamples.* (a) Without latticehood the F_n need not lie in L : this is the corpus’s own Cor. “incompatibility; not a lattice” (`papers/sigma_essential/sigma_essential_body.tex`: — the reviewer marked it consistent-but-not-re-derived (receipt finding 6).

What to look for: open that Corollary once and confirm it states what T3 needs: on the product Ulam carrier the finite intersections of pattern sets escape L . (b) Without countable generation: the countable/co-countable field on ω_1 with the co-countable state — re-derive that the state is σ -additive (a countable disjoint union of countable sets is countable, so the co-countable member is unique) and that the field is genuinely not countably generated (any countable family of countable/co-countable sets generates a field whose atoms include an uncountable set that splits further).

Gap 2.4. *A1(b): blocks closed under countable disjoint unions.* The orthomodular-difference step: for $u = \biguplus c_n$, $c_n \in Bl$, and $b \in Bl$: $u \cap b = \biguplus (c_n \cap b) \in L$, and $u \setminus b$, $b \setminus u \in L$ likewise, so u is compatible with all of Bl , hence in Bl by maximality.

What to look for: each $c_n \cap b \in L$ needs $c_n \leftrightarrow b$ (both in Bl) + L0; the disjointness of $(c_n \cap b)_n$ is inherited; maximality is the Zorn-supplied property from A1(a).

Gap 2.5. *L0's converse (join vs. $\uplus$).* $P \uplus Q \in L$ is an upper bound so $P \vee Q \subseteq P \uplus Q$; the join contains $P \cup Q$ as a set since it bounds each part; equality follows, then $A \cap B = A \wedge B$.

What to look for: this is the step the s12 reviewer singled out and closed; re-run it once — it is four lines and everything concrete-vs-lattice downstream leans on it.

s12 edits to sign off (all cosmetic, listed in the receipt): pentagon irreducibility parenthetical; T1 hypothesis cleaned (every point of $A \cap B$ lies in some member of L inside it — *no exhaustion needed*); corollary remark replaced; P1's use of poorness made explicit; B' (ii) precision caveat (whole-carrier property phrased as coarse-block — must be sharpened before attack).

3 Item 3 (s13 math + s14 edits): attack note §10 — Skeleton A read, theorem-lets R and P, the demotion and factoring

Claim chain: R ((8.1)-inner-regular two-valued f.a. state on Hausdorff = Dirac restriction with compact witness); P (diffuse countable/co-countable components fail (8.1) on uncountable Polish); consequence C (coarse-riding states escape DW Thm D.6 on *every* Polish representation) $\Rightarrow$ Skeleton A demoted, Theorem 2 factors as B' (i) + coarse-block question.

Receipt: ../oml_attack/PROOF_READ_2026-07-10_attack_s10.md (11,686/11,686 checks; sources re-checked verbatim); scripts ../verification/proof_read_2026-07-10_s13/.

Gap 3.1. *R's FIP step for infinite $\mathcal{K}_1$.* The value-1 in-field compacts have the FIP *inside one fixed compact member*; nonempty intersection needs the members closed in that compact (Hausdorff: compact $\Rightarrow$ closed).

What to look for: the s14 rewrite of R's statement (asserting $\mathcal{K}_1 \neq \emptyset$ and $D \neq \emptyset$ as conclusions, avoiding the $\bigcap \emptyset = X$ misparse); then check Hausdorff is genuinely load-bearing via the receipt's counterexample: $\mathbb{N}$ with the cofinite topology, finite/cofinite field, cofinite state — everything is compact, (8.1) trivially holds, no Dirac.

Red flag: any restatement of R that drops Hausdorff or quantifies (8.1) over topological compacts not sandwiched by in-field sets (that is exactly the DW fn-29 gap; see 3.4).

Gap 3.2. *P's topological content* (not scriptable): Cantor–Bendixson on an uncountable Polish space; the perfect kernel is uncountable; compact $\Rightarrow$ closed; and the sub-claim “closed co-countable $\supseteq$ perfect kernel.”

What to look for: re-derive the sub-claim: if F is closed and co-countable, $X \setminus F$ is open and countable, hence contains no point of the perfect kernel (a nonempty open subset of a perfect Polish subspace is uncountable); the receipt’s second counterexample shows “ $x \in$ perfect kernel” is load-bearing ($X = [0, 1] \uplus \{p\}$ isolated, $x = p$ breaks the argument). Reference for background: any DST text (Kechris §6 for Cantor–Bendixson) — also on the skill plan.

Gap 3.3. *P0’s choice flag.* Atom-set countability uses $\text{AC}_\omega(\text{fin})$ on abstract X ; eliminable on Polish X (definable order).

What to look for: confirm P0 is only ever *applied* on Polish carriers in §10 (it is), so the flag is moot in context; nothing to fix, just witness it.

Gap 3.4. *The DW footnote-29 reading.* The claim “DW’s blockwise inner-regularity leg omits $K \in \mathcal{A}_i$, gap confirmed in the text, no repair in Appendix D” is a *source-reading* judgment.

What to look for: Derr–Williamson 2023, arXiv:2302.03522 (derr_williamson_2023.pdf): Thm D.6 p. 48 (five legs; blockwise inner regularity is a distinct leg), footnote 29 (verbatim, membership-free), proof p. 49 (“all restrictions μ_i are inner regular” — no repair). Then check the corpus convention: we cite D.6 only in the (8.1) sense (Maharam-style in-field sandwich), which is the reading that makes their proof work.

Red flag: citing D.6 with fn-29’s literal (membership-free) reading anywhere — the s13 Addendum exists precisely to pin this.

Gap 3.5. *Maharam anatomy M1–M3* (source-reading; verbatim anchors already in the receipt).

What to look for: Maharam 1972 (maharam_1972.pdf): (8.1) display p. 145 (with $K \in \mathcal{F}_\alpha$ — the in-field membership DW’s footnote drops); the Hahn–Banach engine is her *Lemma* 6.1 (Thm 6.1 is the criterion invoking it); her Remark p. 146 (the m_α “have countably additive extensions”). Her proof of 8.1 (p. 146) uses (8.1) only for the compact/open sandwich and the $K \subseteq G_1 \cup \dots \cup G_n$ step — this is what licenses §10’s “(8.1) consumed ONLY as intersection-stable compact class.”

Gap 3.6. *The all-Hausdorff strengthening* (banked s14): diffuse mass $c > 0$ kills (8.1) on *every* Hausdorff topology, by reduction to R.

What to look for: the reduction: an in-field compact $K \subseteq F$ with $\mu(K) > \mu(F) - c$ must be co-countable, so the co-countable component ν would itself satisfy (8.1), contradicting R (ν non-principal). The threshold arithmetic is machine-checked; the topological conclusion is the hand part.

s14 edit to sign off: the scope fix on “fails *exactly* on the coarse locus” — earned only among *coarse-block* σ -states (atomic states satisfy (8.1) there via finite in-field atom-truncations; compact-poor blocks fail (8.1) even at Diracs: $\delta_\omega \upharpoonright \{\emptyset, \text{irr}, \mathbb{Q}, \mathbb{R}\}$, ω irrational). Propagated to §10d(i) and the prior-art verdict banner. The s13 conclusions use only the failure direction and are unaffected — verify that one sentence in the receipt and you have verified the propagation’s safety.

4 Item 4 (s15 math + s16 edits): attack note §11 — the $B'(i)$ reduction ladder and the Marczewski read

Claim chain (as reviewed, C1–C12): 2BR two-block rescue; cluster normal form; $P^\perp$ (pointed $\Leftrightarrow \sigma$ on countably generated blocks); kernel-FIP strictly stronger than coherence; representation-dependence of $K(s) = \emptyset$; $\Phi \Leftrightarrow \text{St}_\sigma$ dense in St_{fa} ; in-block 4(i) + compact-transport criterion; ω_1 scoping; no-cheap-union; T4 ladder.

Receipt: ../oml_attack/PROOF_READ_2026-07-11_attack_s11.md (424,509/424,509 checks); scripts ../verification/proof_read_2026-07-11_s15/; Marczewski source-check inside the receipt.

Gap 4.1. $P^=$'s *infinite legs* (scripts cover only the finite pentagon). Three joints: (a) the $(\Rightarrow)$ leg is T3 run on a countable generating family — inherits item 2's gap 2.2; (b) the kernel-nonempty $\Rightarrow$ charges-an-atom bridge: for $\omega \in D_{Bl}$, if $\nu(A(\omega)) = 0$ then $\nu(A(\omega)^c) = 1$ forces $\omega \in A(\omega)^c$, absurd — needs C4's atomicity; (c) the $(\Leftarrow)$ leg: $\omega \in D_{Bl}$ makes $\nu \upharpoonright Bl = \delta_\omega \upharpoonright Bl$ (two-valued + field), which is σ -additive per block, then A1c globalizes.

What to look for: C4's "every member of Bl is a union of signature cells": the family of cell-saturated sets is a σ -field containing the generators — write the countable-union line once. Note the banked sharpening: $(\Leftarrow)$ needs *no* countable generation.

Red flag: any later use of $P^=$ on a block *not* countably generated in the $(\Rightarrow)$ direction — that direction is exactly what coarse blocks refuse (ω_1 counterexample).

Gap 4.2. Φ -density's topology (C8). Three hand steps: (a) $St_{fa} \subseteq \{0,1\}^L$ is closed — each finite-additivity constraint involves finitely many coordinates, so violations are witnessed on basic clopens; (b) compactness of $\{0,1\}^L$ is Tychonoff (BPI-strength — fine in ZFC, flagged for the record); (c) the pattern-vs-clopen correspondence: a finite value assignment with an f.a. witness $\perp \uplus$ -closes to a finite $\perp$ -closed pattern *without leaving the clopen* (the closure only adds forced values). The map pattern $\mapsto$ trace is onto, not injective — harmless, but the "correspond exactly" phrasing should be read as "nonempty traces."

What to look for: then the equivalence itself: Φ says every nonempty basic clopen of St_{fa} meets St_σ — literally density. Check Φ 's quantifier in papers/sigma_essential/sigma_essential_body.tex (§Discussion) is over *finite* patterns, which it is.

Gap 4.3. The ω_1 *scoping bank* (C10). (a) The co-countable family (with $\emptyset$ adjoined — s16 fix) is a countably compact class: a countable intersection of co-countables is co-countable, nonempty since ω_1 is uncountable (uses AC_ω : countable unions of countable sets); any sequence containing $\emptyset$ vacuously fails the FIP hypothesis. (b) It Marczewski-approximates the countable/co-countable field w.r.t. the coarse state: co-countable E takes $D = K = E$; countable E takes $K = \emptyset$. (c) Independent corroboration: the state is purely atomic (ω_1 one atom), hence compact by M–RN §3(ii) directly.

What to look for: Marczewski, "On compact measures," *Fund. Math.* 40 (1953) 113–124 (marczewski_1953_on_compact_measures.pdf): compact class §2 p. 115 (sequence form; purely set-theoretic — his stated aim p. 113 is "eliminating non-essential topological concepts"); approximation §3 p. 116 (η -form over *all* $E \in \mathbf{M}$ — this is why $\emptyset$ is needed); compact measure §4 p. 118; 4(i) p. 118.

Red flag: treating Marczewski compactness as if it implied topological (8.1)-regularity — the whole point of the bank is that the coarse state is Marczewski-compact yet kills the topological leg.

Gap 4.4. The s16 wording *fixes* — these change standing conjecture targets; sign off explicitly. (a) Compact-transport criterion and the coarse-factor wall statement now read "Marczewski-approximates in-block" instead of "refines the value-1 filter." The two are *inequivalent*: the class of all singletons of Ω is countably compact and refines every ultrafilter, which would make every two-valued f.a. state σ -additive — absurd (any nonprincipal ultrafilter state on $\mathcal{P}(\mathbb{N})$). The value-1 *inner* witness $D \in Bl$, $\mu(D) = 1$, $D \subseteq K$ is what powers the FIP step.

What to look for: re-run the criterion’s proof once with the corrected hypothesis: tails F_n value-1, sandwiches $D_n \subseteq K_n \subseteq F_n$, $\mu(D_n) = 1$; finite intersections of the D_n are value-1 (in-block ultrafilter), hence nonempty, giving the K_n the FIP; countable compactness yields $\emptyset \neq \bigcap K_n \subseteq \bigcap F_n = \emptyset$. The coarse-block toy hunt will aim at the wall statement in this corrected form — this is the one edit with forward-pointing consequences. (b) 5(iii)/5(iv): the clean “independent σ -fields + compact partials $\Rightarrow$ compact” is 5(iv) (p. 120); 5(iii) puts its hypotheses on the *approximating classes* and partial compactness alone is explicitly insufficient there (“and, what is more”). Both are compactness-*transfer* theorems, not extension-existence (that is Marczewski 1951, *Fund. Math.* 38, Thm I). (c) M–RN §4 credit: the positive half (minimal σ -extensions of compact measures are compact) is M 1953 4(ii), quoted in M–RN as “C 4 (ii)”; M–RN §4’s own theorem is the converse-false example. Reference: marczewski_ryll_nardzewski_1953.compactness.products.pdf, pp. 165–170.

Gap 4.5. *The Zorn frame and the crux scope* (C12(iii) + s16 scope note). T4’s *necessity* for $B'(i)$ is clean ($B'(i) \Rightarrow \sigma$ -extension of the cluster-pattern $\Rightarrow$ pointed at every block by $P^\perp \Rightarrow$ the charged atom of Bl_0 witnesses coherence). But the boxed crux is at *finite* support, and the Zorn run needs the successor at arbitrary $\mathcal{S}$: a block may have infinitely many atoms, so “point Bl_0 somehow” is an infinite union of closed conditions, not closed — compactness alone does not close the limit stage.

What to look for: the s16 annotation at the “everything reduces” sentence; confirm you are happy with the honest status: T4 is a genuine *falsification* target (a T4-violating configuration kills $B'(i)$ outright), while the *proof* direction needs T4 + a completion principle still to be found.

Red flag: any future note claiming “ $T4 \Rightarrow B'(i)$ ” unconditionally.

Gap 4.6. *Small remainders* (from the receipt’s gap list): C11’s empty *total* intersection is a limit statement (script checks the finite shadow — the two-class witness on $\mathbb{N}$ is three lines by hand); the general- Ω halves of C1/C1'/C2/C3 are the written proofs in §11a–b (each under ten lines; the machine corroborated all finite instances); the monotonicity strengthening (any concrete σ -class, two lines: $E \subseteq A$, $\mu(E) = 1$, $\mu(A) = 0$ gives $\mu(E \uplus A^c) = 2$).

5 Item 5 (s7 + s10, reconstruction lane): Theorem P, Lemma NG, Theorem B (the pruning stack)

Claim chain: Theorem P (exact characterization, all k , all L , no strong-connectivity hypothesis: $TR_k(\text{rot-1}) = \text{LISC}_k$); Lemma NG (non-generator rotations overcount: orbit decomposition to $\text{LISC}_{k/d}$); Theorem B (effective certificate: k -cap, first-repeat minimality, union bookkeeping) + the Safe(ρ) instrument.

Receipt: papers/reconstruction/notes/PROOF_READ_2026-07-10.pruning.md; proof papers/reconstruction/notes/scripts papers/reconstruction/oracles/proof_read_2026-07-10/. Machine coverage is unusually strong here ($k \leq 8$ exhaustive on adversarial targets, including non-strongly-connected and multi-component); the hand gaps are the general- k arguments.

Gap 5.1. *Step 3 (assembly $\Leftarrow$) at general k .* The walks re-assembled from $TR^{(1)}$ witnesses are simple, of length kL , ring-consistent — verified computationally on five targets chosen to tempt collisions, but the general argument is the written proof’s.

What to look for: the simplicity argument: layer-injectivity (Lemma 0’s layer clock) is what forbids revisits; check the proof never assumes strong connectivity (the receipt confirms the

oracle cross-check included non-SC and multi-component graphs, so the *statement* is right; your check is that the *written argument* also avoids it).

Gap 5.2. *Step 4 (generator conjugation) at general k .* The identity $P(\sigma_1(u))[j] = \sigma_r(P(u))[j]$ reduces to $r\bar{r} \equiv 1 \pmod{k}$; machine-checked for all 21 pairs with $k \leq 8$.

What to look for: the general congruence argument is three lines of modular arithmetic in the written proof — verify the index bookkeeping once (this is where an off-by-one would hide; the machine found none up to 8).

Gap 5.3. *Theorem $B(1)$ ’s pigeonhole and the $k \geq 2$ scoping.* Unsafe is a union over $2 \leq k \leq |A|$; the k -cap is pigeonhole + Theorem P.

What to look for: the union bounds match the instrument’s **range(2, A+1)** (receipt confirms); the $k = 1$ remark in the proof is inert reader-reassurance (receipt finding 2) — nothing to verify beyond noticing it claims nothing.

Gap 5.4. *Citations flagged non-load-bearing.* Wielandt / Schwarz / Dulmage–Mendelsohn are bridge references only; the brute repeat-search is what is certified.

What to look for: confirm the proof text flags them as such (it does per the receipt); no action unless you want the classical anchors verified for the eventual write-up. Cosmetic: instrument targets $\text{adv2} \cong \text{adv3}$ (relabeling), so “15 targets” is 14 distinct — worth a one-word edit only if the instrument log is ever quoted.

After ratification

Tell me the outcome per item (ratify / ratify-with-notes / veto), and I will: mark the taxonomy rows and MEMORY hooks RATIFIED with date; strike the “awaits user ratification” banners in the attack note, frontier map, and shovel plan; and (for any veto) open the repair as the next session’s first item. The forward queue after that is unchanged: T4 via Maharam (8.2), coarse-block toy hunt (target: the corrected wall statement), B' (ii) restatement, PP1994; parallel relaxed-gate items FIDELITY_REVIEW, abstract call, statistics verdict, ILL hunt.